A three-parameter family for the Erdős–Straus equation and its computational reach

Working notes, 8 November 2026

Status. The Erdős–Straus conjecture ($4/n = 1/x + 1/y + 1/z$ for all $n \geq 2$, Erdős Problem 242) is open and remains open after this note. Everything below is labelled PROVED, VERIFIED (with its bound) or CONJECTURED.

1 An explicit identity

Theorem 1 (PROVED)**.** *Let m, u, t be positive integers with*

$$n \;=\; mut - m - u \qquad \textit{and} \qquad mt \equiv 1 \pmod 4.$$

Put $w = (mt-1)/4$ and $X = ut - 1$. Then

$$\frac{4}{n} = \frac{1}{uw} + \frac{1}{n\,w\,X} + \frac{1}{u\,w\,X}.$$

Proof. Since $mt \equiv 1 \pmod 4$, w is a positive integer. From $n + m + u = mut$ we get $n + u = m(ut-1) = mX$. Then

$$\frac{1}{u} + \frac{1}{nX} + \frac{1}{uX} = \frac{nX + u + n}{unX} = \frac{n(X+1) + u}{unX} = \frac{nut + u}{unX} = \frac{nt+1}{nX}.$$

Expanding, $(mt-1)(ut-1) = mut^2 - mt - ut + 1 = t(mut - m - u) + 1 = nt + 1$, so $nt + 1 = 4wX$. Hence

$$\frac{1}{uw} + \frac{1}{nwX} + \frac{1}{uwX} = \frac{1}{w} \cdot \frac{nt+1}{nX} = \frac{1}{w} \cdot \frac{4wX}{nX} = \frac{4}{n}. \qquad \square$$

Remark. If $m \equiv t \equiv 3 \pmod 4$ then $mt \equiv 1 \pmod 4$ and $n \equiv -m \equiv 1 \pmod 4$ is forced, so the family concerns $n \equiv 1 \pmod 4$. The identity does not need n prime.

2 Reformulation and known special cases

Write $x = (n+m)/4 = uw$. The identity is the divisor criterion "$m/(nx) = 1/y + 1/z$ is solvable iff some $d \mid (nx)^2$ satisfies $d \equiv -nx \pmod m$" evaluated at $d = n^2 w$. Equivalent forms of the hypothesis are

$$n + m + u = mut, \qquad (mt-1)(ut-1) = nt + 1, \qquad n \equiv -m \pmod{mt-1}.$$

Proposition 1 (PROVED)**.** *For a prime $p \equiv 1 \pmod 4$:*

1. *If $u = 1$, the family holds exactly when $p + 1$ has a divisor $m \equiv 3 \pmod 4$. This is the classical criterion of Obláth (listed on erdosproblems.com/242).*

2. *In the divisor criterion, writing $d = p^i e$ with $e \mid x^2$, the choice $e = x^2$ (with $i = 0$ or 2) never works at any offset: it would need $m \mid p^2 + 4$ with $m \equiv 3 \pmod 4$, but every odd divisor of $p^2 + 2^2$ is $\equiv 1 \pmod 4$ because $\gcd(p, 2) = 1$.*

Proof. (1) With $u = 1$ the equation is $p+1 = m(t-1)$; if $m \equiv 3 \pmod 4$ divides $p+1 = 2 \cdot o$ then $t - 1 = 2o/m$ and $t \equiv 3 \pmod 4$ holds because o/m is odd. For (2): a prime $q \mid p^2 + 4$ with q odd satisfies $-4 \equiv p^2$, so -1 is a square mod q, so $q \equiv 1 \pmod 4$. □

Proposition 2 (PROVED; divisor form)**.** *Let $p \equiv 1 \pmod 4$. Then p lies in the family of Theorem 1 if and only if there are an integer $t \equiv 3 \pmod 4$ and a divisor a of $pt + 1$ with $4 \mid a$ and $a \equiv -1 \pmod t$ (and $b := (pt + 1)/a$, which then satisfies $b \equiv -1 \pmod t$, is automatically admissible).*

Proof. ($\Rightarrow$) Take $a = mt - 1 = 4w$ and $b = ut - 1 = X$; the identity $(mt - 1)(ut - 1) = pt + 1$ was shown in the proof of Theorem 1. ($\Leftarrow$) From $ab = pt + 1 \equiv 1 \pmod t$ and $a \equiv -1$ we get $b \equiv -1 \pmod t$. Set $m = (a + 1)/t$, $u = (b + 1)/t$. Then $(mt - 1)(ut - 1) = pt + 1$ expands to $mut^2 - mt - ut = pt$, i.e. $mut - m - u = p$, and $mt = a + 1 \equiv 1 \pmod 4$. □

Thus the conjecture for $p \equiv 1 \pmod 4$ would follow from: *for every such prime there is $t \equiv 3 \pmod 4$ such that $pt + 1$ has a divisor $a \equiv -1 \pmod t$ with $4 \mid a$.* For $t = 3$ this says $3p + 1$ has a divisor $\equiv 11 \pmod{12}$ divisible by 4, and so on; each t imposes a different multiplicative condition, and no single t works for all p.

3 Computations

All searches use $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ (the classes Mordell's identities leave open). For each such prime the code searches $u = 1, 2, \ldots$ for a divisor $m \equiv 3 \pmod 4$ of $p + u$ with $(p + u)/m \equiv -1 \pmod u$ and $t = ((p + u)/m + 1)/u \equiv 3 \pmod 4$.

Bound	Hard primes	Largest minimal u
$p < 6 \cdot 10^5$	1469	161 (at $p = 30241$, $(m, t) = (27, 7)$)
$p < 2 \cdot 10^7$	39391	890 (at $p = 5101441$, $(m, t) = (39, 147)$)

VERIFIED, $p < 2 \cdot 10^7$: every hard prime has a hit with $u \leq 890$, and the triple of Theorem 1 equals $4/p$ in exact rational arithmetic for all 39391 primes. The share of primes with minimal $u = 1, 2, 3$ is $43\%, 23\%, 13\%$.
VERIFIED, $p < 2 \cdot 10^6$ (4519 hard primes), via the independent t-search of Proposition 2: every prime has a hit with $t \leq 367$ (worst: $p = 830041$, $(t, m, u) = (367, 87, 26)$); the control $p = 7 \equiv 3 \pmod 4$ correctly returns no hit.
VERIFIED, random sample (seed 20261108): 40 random primes in the six open classes at each of 12, 15 and 18 digits all lie in the family, with minimal $u \leq 12$, 8, 5 respectively; the control $p = 1000003 \equiv 3 \pmod 4$ returns no hit. This is a sample, not a range.
Controls: replacing x by $x+1$ breaks the identity for 39391 of 39391 primes; feeding the independent divisor-criterion solver the wrong offset $m + 4$ or $m + 8$ fails for 55265 of 78782 attempts (the remainder succeed legitimately at another offset).

4 What is and is not established

- PROVED: Theorem 1; the $u = 1$ characterization; the failure of the $e = x^2$ route.

- VERIFIED, $p < 2 \cdot 10^7$: coverage of the six open classes by the family with $u \leq 890$.
- CONJECTURED: every prime $p \equiv 1 \pmod 4$ lies in the family; the minimal u stays below a power of $\log p$. Neither is proved, and the family is a cubic-surface integral-point problem of Markoff type, which is the setting where Bright and Loughran showed there is no Brauer–Manin obstruction but integral points are not known to exist.
- Not established: ES for all primes; any statement about the u needed for general p.
- Prior art: the criterion and the $u = 1$ case are classical. Comparison with Chamberland's representation $p = qr - 4s_1s_2$ ($q \equiv 3 \bmod 4$, $s_i \mid (q+1)/4$): VERIFIED, odd primes $p \equiv 1 \pmod 4$ below 400, the two families are *not* the same representation-by-representation (e.g. $p = 17$ has $q = 7$ but no family member with $m = 7$; $p = 37$ has $m = 3$ but no Chamberland $q = 3$), although both are non-empty for all 36 primes tested. Whether they are equivalent as covering statements is not established. I did not check the Elsholtz–Tao Type II parametrization symbol by symbol, so no priority is claimed.